

YOLO11n vs ESPDet-Pico: Comparative Analysis

This section presents a controlled comparison between standard YOLO11n and the ESP-optimized ESPDet-Pico architecture. Both models were trained with identical configurations to isolate the impact of architectural modifications for edge deployment.

Experimental Setup

To ensure a fair comparison, both models were trained under identical conditions:

- **Dataset:** Grape leaf detection dataset (train/val/test splits)
- **Image size:** 320×416 pixels (rectangular training)
- **Batch size:** 32
- **Epochs:** 1000 (early stopping with patience=75)
- **Optimizer:** AdamW (lr=0.002, weight decay=0.0005)
- **Augmentation:** HSV (h=0.015, s=0.4, v=0.4), rotation=10°, scale=0.5
- **Pretrained weights:** yolo11n.pt (same initialization)

Detection Performance Comparison

Table 1 presents the detection performance metrics for both models on the test dataset. All metrics are evaluated using deployment-matching parameters (confidence threshold=0.25, IoU threshold=0.7, max detections=10).

Table 1: Detection performance comparison between YOLO11n and ESPDet-Pico

Metric	YOLO11n	ESPDet-Pico	Difference	Winner
mAP@50 (%)	61.75	56.42	−5.33	YOLO11n
mAP@50-95 (%)	43.57	32.29	−11.28	YOLO11n
Precision (%)	69.55	62.77	−6.78	YOLO11n
Recall (%)	57.29	54.17	−3.13	YOLO11n
Best Epoch	327	711	+384	YOLO11n

Model Size and Deployment Characteristics

Table 2 compares the model sizes and deployment characteristics. The significant size reduction in ESPDet-Pico enables deployment on resource-constrained ESP32-S3 micro-controllers.

Table 2: Model size comparison and deployment characteristics

Property	YOLO11n	ESPDet-Pico	Reduction
FP32 Model Size (MB)	5.23	1.06	79.8%
INT8 Model Size (KB)	—	478	—
Deployable on ESP32-S3	No	Yes	—
Memory Footprint (PSRAM)	>8 MB	492 KB	—
Inference Time (ms/frame)	—	366	—

Analysis and Discussion

The comparison reveals a trade-off between detection accuracy and deployment feasibility:

1. **Accuracy Trade-off:** ESPDet-Pico achieves 56.42% mAP@50, which is 5.33 percentage points lower than YOLO11n’s 61.75%. This represents a modest accuracy reduction of 8.6% relative to the baseline.
2. **Model Size Reduction:** ESPDet-Pico demonstrates an impressive 79.8% reduction in model size (1.06 MB vs 5.23 MB for FP32 weights). After INT8 quantization, the model further compresses to 478 KB, making it deployable on ESP32-S3 devices with 8 MB PSRAM.
3. **Deployment Feasibility:** Unlike standard YOLO11n, which exceeds ESP32-S3 memory constraints, ESPDet-Pico successfully deploys and operates within the 8 MB PSRAM budget. The model achieves 366 ms inference time per frame on the ESP32-S3 (dual-core Xtensa LX7 @ 240 MHz).
4. **Training Efficiency:** ESPDet-Pico converged at epoch 711, significantly later than YOLO11n (epoch 327). This suggests that the ESP-optimized architecture requires more training iterations to reach optimal performance.
5. **Precision vs Recall:** ESPDet-Pico maintains relatively high precision (62.77%) compared to recall (54.17%). In the context of a two-stage detection-classification pipeline, high precision is critical to minimize false positives and reduce wasted bandwidth in LoRaWAN transmission.

Conclusion

The 5.33% accuracy reduction in ESPDet-Pico represents an **acceptable trade-off** for enabling edge deployment on ESP32-S3 microcontrollers. The 80% model size reduction and successful INT8 quantization to 478 KB demonstrate that architectural optimization for edge devices can maintain reasonable detection performance while enabling deployment on severely resource-constrained platforms. For precision agriculture applications where edge inference is essential, ESPDet-Pico provides a practical solution that balances accuracy with deployability.

Key Findings

- ESPDet-Pico achieves **56.42% mAP@50** with **80% smaller model size**
- Accuracy reduction of **5.33 percentage points** enables ESP32-S3 deployment
- Post-quantization model (**478 KB INT8**) fits within 8 MB PSRAM constraint
- Real-time inference achieved: **366 ms per 416×320 frame**
- High precision (**62.77%**) minimizes false positives in two-stage pipeline